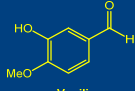




## 第十章 醛和酮

### Aldehydes and Ketones



Vanillin  
香草醛 (香兰素)



Camphor  
樟脑

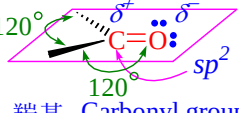
1

### 醛酮结构

醛、酮的官能团是羰基，羰基碳原子为 $sp^2$ 杂化，带部分正电荷，是极性官能团

$\text{R}-\text{C}(=\text{O})-\text{H}$   
醛

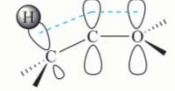
$\text{R}-\text{C}(=\text{O})-\text{R}$   
酮



羰基 Carbonyl group

**Acidic  $\alpha$ -hydrogens**

 $\text{H} \rightarrow \text{C} \rightarrow \text{C} \Rightarrow \text{O}$   
 羰基的吸电子诱导效应



$\alpha$ 碳氢键对羰基的超共轭效应

2

➤ 广泛存在于自然界中, 可从动植物体内提取

柠檬醛、薄荷酮、樟脑、麝香

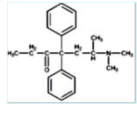
雄酮激素、雌酮激素、可的松、炔诺酮、视黄醛

**应用**

➤ 有机合成的重要原料、溶剂，如丙酮、乙醛等；

➤ 香料、香精，如月桂醛、柠檬醛、茴香醛等；

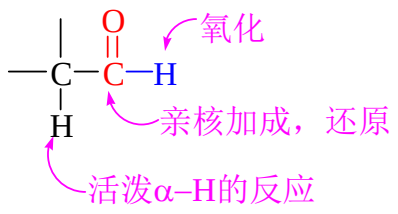
➤ 药物，如美沙酮（戒毒）、盐酸麻黄苯丙酮（冠心病）等。



3

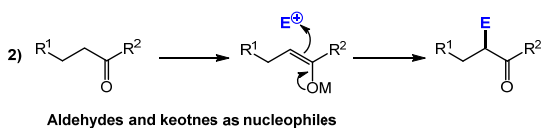
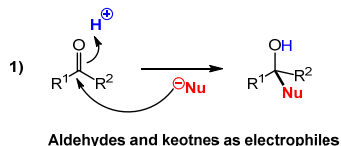
### 醛酮的亲核加成反应

#### 醛酮的反应概貌



4

## Reactions of Aldehydes and Ketones



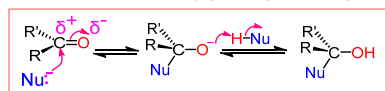
3) Oxidation reactions

结构决定性质 (反应)

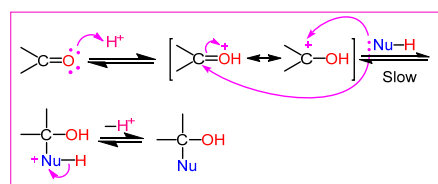
5

## 普遍的反应机理

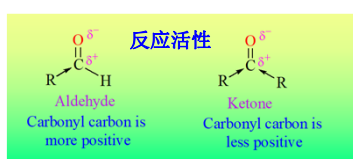
### 1. Addition of a strong (basic) nucleophile



### 2. Acid-catalyzed nucleophilic addition



6



醛 > 脂肪酮 > 芳香酮; 环酮 > 脂肪酮

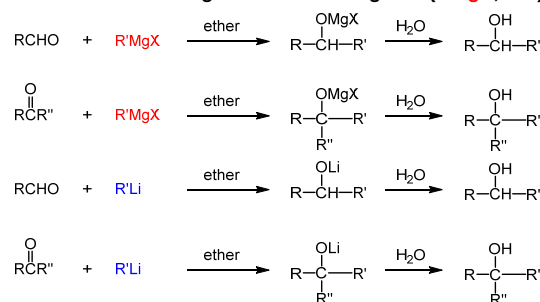
### 亲核试剂的分类

- 含碳的亲核试剂:  $\text{RMgX}$ ,  $\text{HCN}$ ,
- 含氧的亲核试剂:  $\text{H}_2\text{O}$ ,  $\text{ROH}$
- 含硫的亲核试剂:  $\text{NaHSO}_3$
- 含氮的亲核试剂:  $\text{RNH}_2$ ,  $\text{R}_2\text{NH}$ ,  $\text{H}_2\text{N-G}$

7

## 羰基与含碳亲核试剂的加成

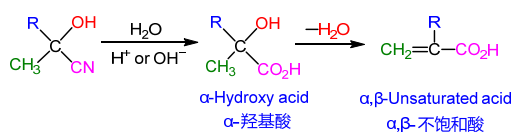
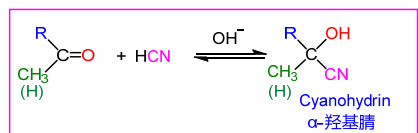
### 1. The Addition of Organometallic Reagents ( $\text{RMgX}$ , $\text{RLi}$ )



制备一级醇: 多聚甲醛 (Paraformaldehyde)

8

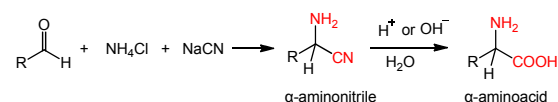
## 2. The Addition of Hydrogen Cyanide (HCN)



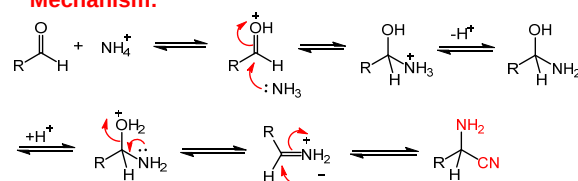
9

## Strecker反应

羰基化合物与氯化铵、氰化钠反应，形成α-氨基腈，经水解可以制备α-氨基酸：

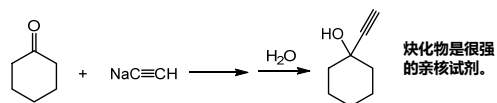


### Mechanism:

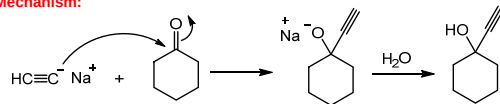


10

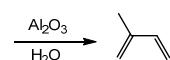
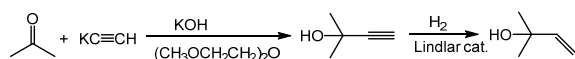
## 3. The Addition of Acetylide (炔化物)



### Mechanism:



### Example:

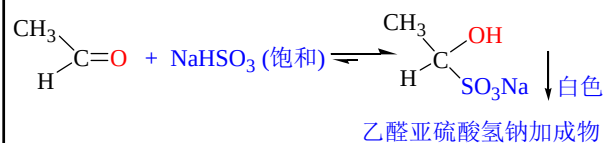


异戊二烯的工业合成路线

11

## 羰基与含硫亲核试剂的加成

### 与亚硫酸氢钠的反应

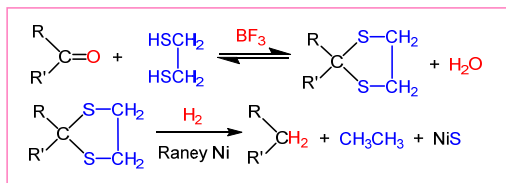


- 醛、脂肪族甲基酮、C8以下环酮能发生反应，其他酮不反应
- 本反应可用于鉴别和分离提纯

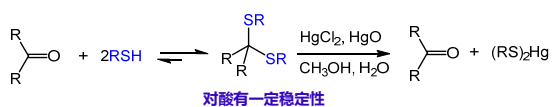
12

## 2. The Addition of Thiol (RSH)

强亲核性



脱除醛酮的羰基氧 (还原)



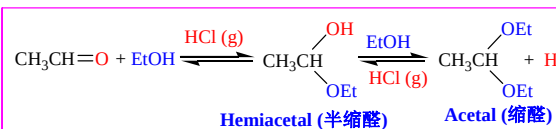
对酸有一定稳定性

保护醛酮的羰基

13

## 羰基与含氧亲核试剂的加成

与醇的加成——形成半缩醛(酮), 缩醛(酮)



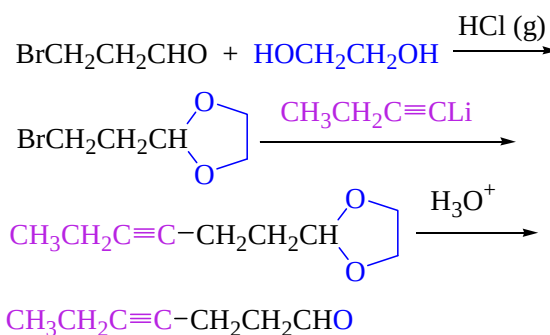
14

## 反应特点

- 半缩醛(酮)无论在酸性或碱性介质中均不稳定
- 缩醛(酮)在碱性介质中稳定, 在酸性介质中不稳定
- 缩醛较易形成, 缩酮较难
- 用乙二醇形成环状缩酮反应可提高缩酮的产率

15

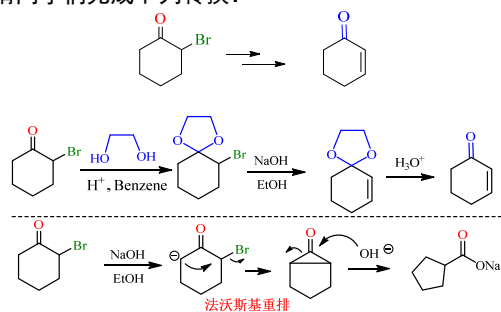
## 应用: 保护羰基



16

## 测试

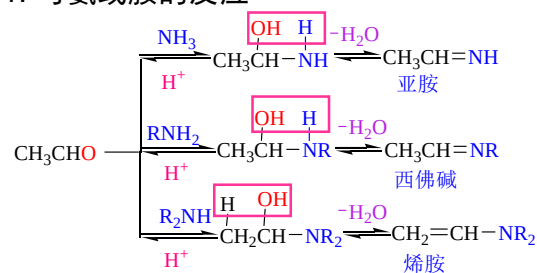
请同学们完成下列转换：



17

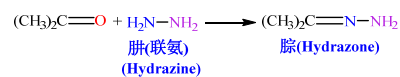
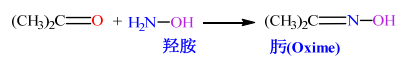
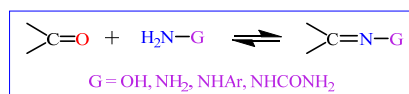
## 羰基与含氮亲核试剂的加成

## 1. 与氨或胺的反应 可逆反应，分水操作



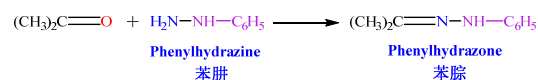
脂肪族Schiff 碱不稳定,芳香族Schiff 碱稳定

18



醛和非对称酮的产物构型如何？ E型产物为主

19

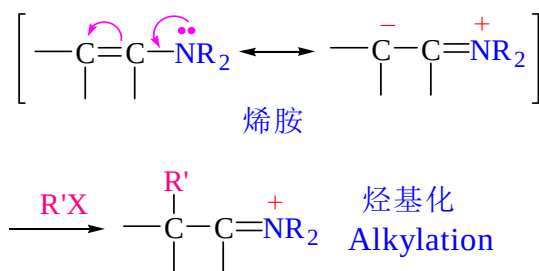


醛肟显色剂：

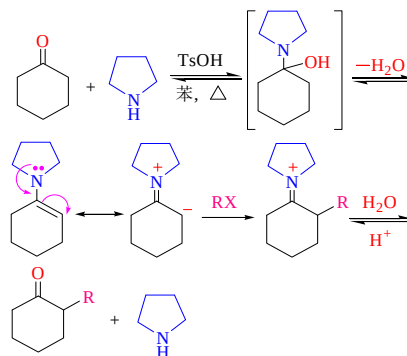


20

## 烯胺的性质及应用



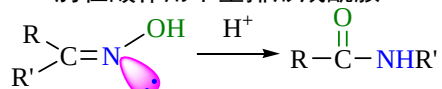
21



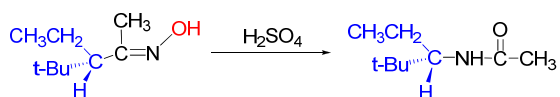
22

## Beckmann 重排

肟在酸作用下重排形成酰胺

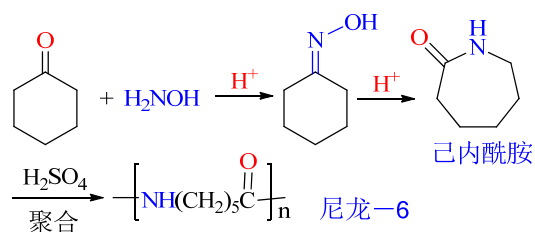
常用催化剂:  $\text{H}_2\text{SO}_4$ ,  $\text{HCl (g)}$ ,

- 迁移基团处于羟基的反式位置
- 迁移基团如果是手性碳原子, 在迁移过程中其构型保持不变



23

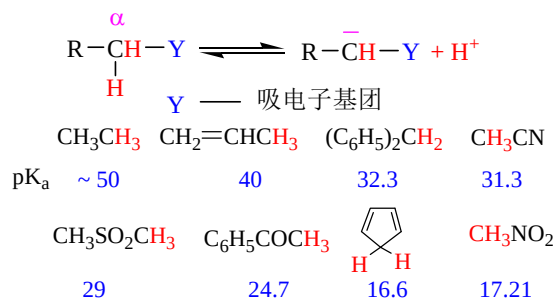
## 应用：由酮制备酰胺



24

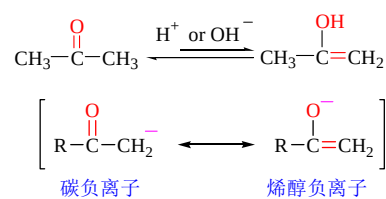
## $\alpha$ -活泼氢的反应

### 1. $\alpha$ -氢的活性

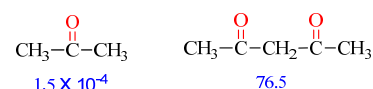


25

## 酮式和烯醇式——互变异构



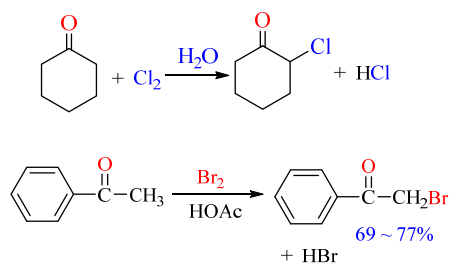
烯醇式  
含量



26

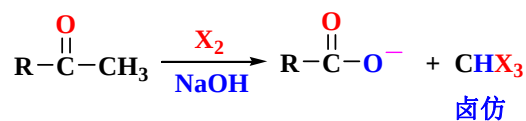
## $\alpha$ -活泼氢的反应

### 2. 卤化和卤仿反应



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## 卤仿反应

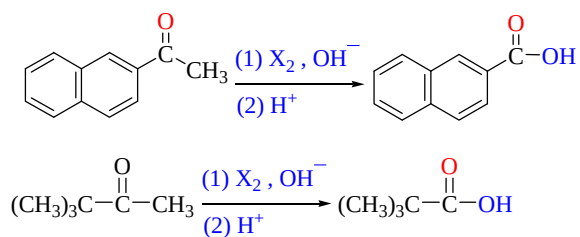


具有  $\text{—}\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_3$     $\text{—}\overset{\text{OH}}{\underset{|}{\text{CH}}}-\text{CH}_3$  结构的  
化合物都能发生反应。碘仿反应可用于鉴别。

碘仿  $\text{CHI}_3 \downarrow$  黄色

28

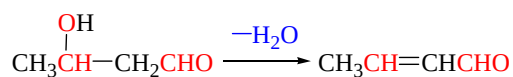
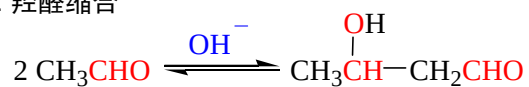
## Example



29

 $\alpha$ -活泼氢的反应

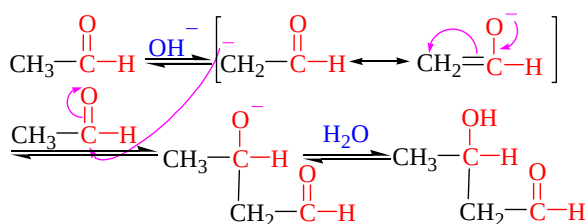
## 3. 羟醛缩合



- 含有 $\alpha$ -H的醛(酮)在稀碱的作用下,缩合生成 $\beta$ -羟基醛(酮)
- $\beta$ -羟基醛(酮)易脱水形成 $\alpha, \beta$ -不饱和醛(酮)

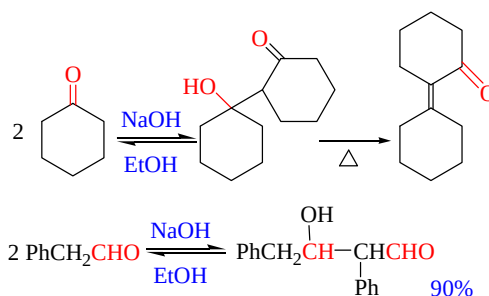
30

## 反应机理 (掌握)



31

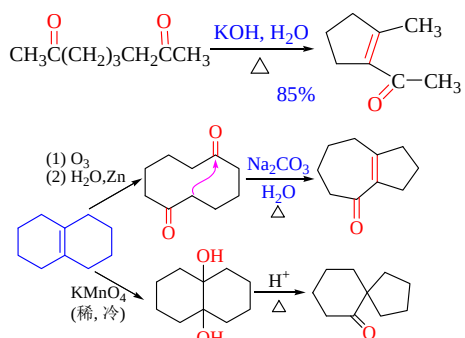
## 分子间缩合



32



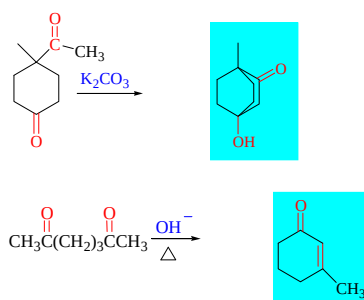
## 分子内缩合



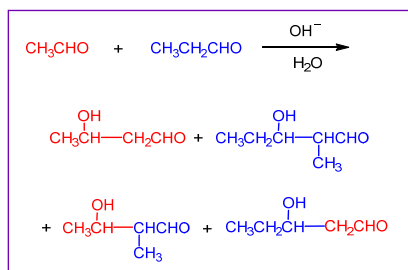
33

## 测试

请同学们完成下列反应



34

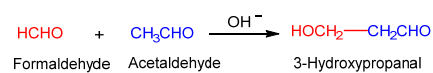
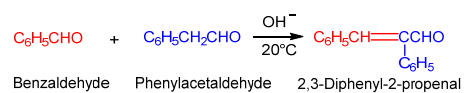
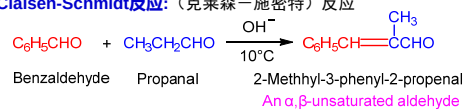
2. Cross aldol additions/condensations  
交叉缩合

多种产物生成，包括相应的消除产物 (aldol condensation)

35

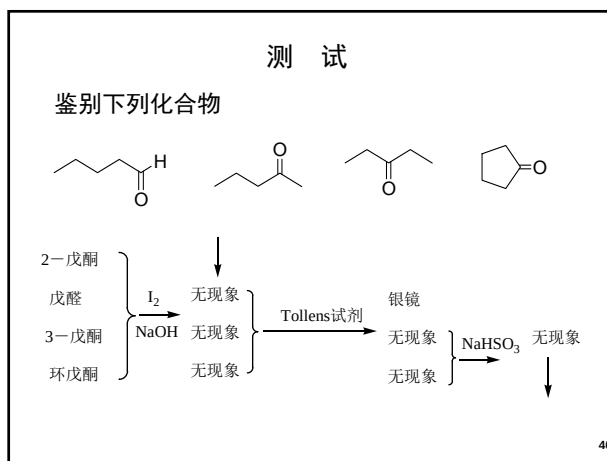
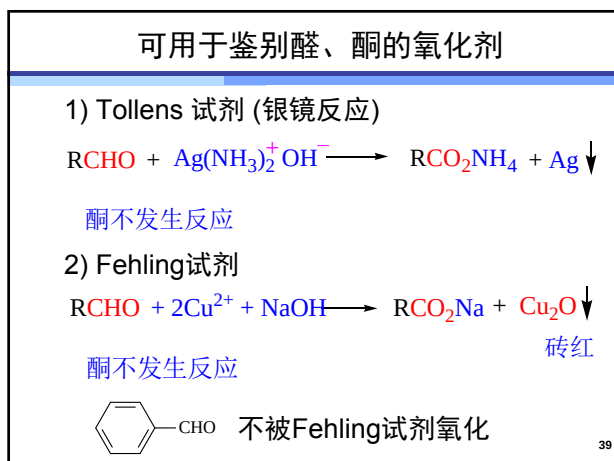
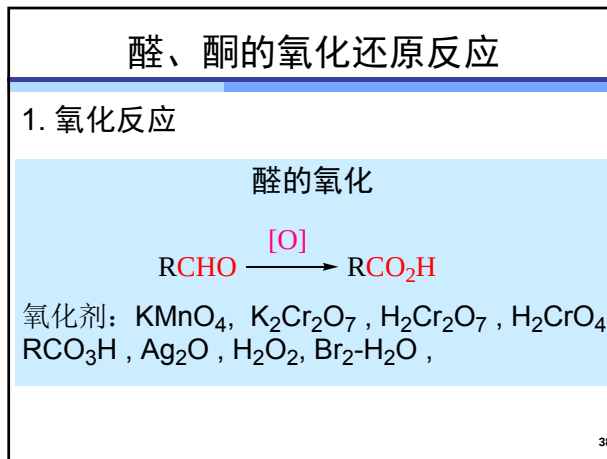
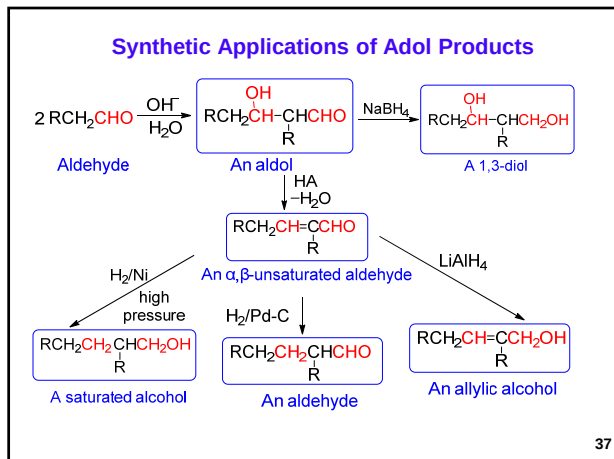
## Practical Cross Aldol Reactions

Claisen-Schmidt反应: (克莱森-施密特) 反应



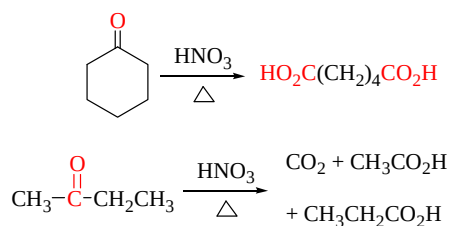
反应的选择性

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### 酮的氧化

- 酮遇一般氧化剂时，不被氧化
- 酮遇强氧化剂时，发生碳链断裂，氧化成酸

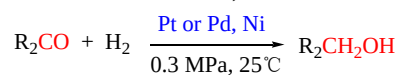
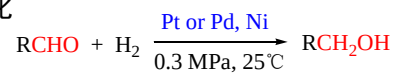


41

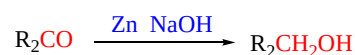
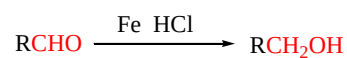
### 醛、酮的还原反应

#### 2. 还原为羟基

##### 催化氢化



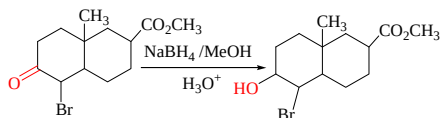
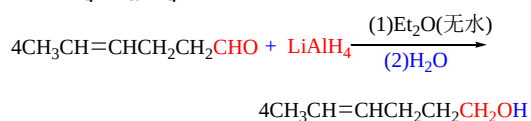
##### 金属还原剂



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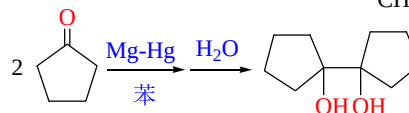
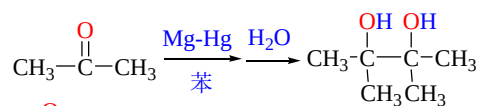
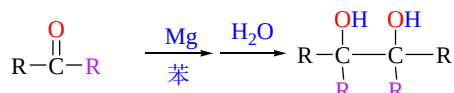
### 用金属氢化物还原

$\text{LiAlH}_4$   $\text{NaBH}_4$



43

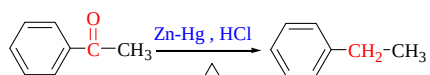
### 双分子还原



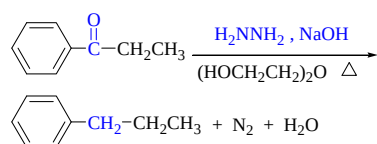
44

还原为亚甲基或甲基

### (1) Clemmensen Reduction



## (2) Wolff-Kishner-黄鸣龙还原

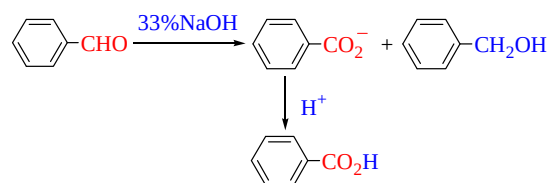


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## 岐化反应 Cannizzaro 反应

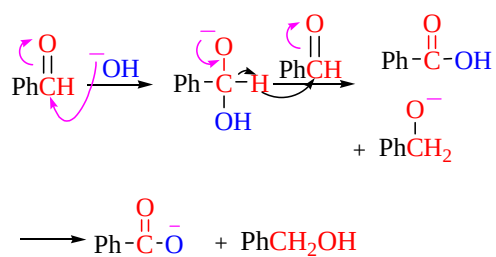
岐化反应 Cannizzarro reaction

不含 $\alpha$ -H的醛在浓碱作用下发生分子间的氧化-还原反应生成相应的醇和酸



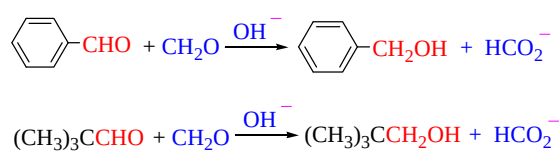
46

### 反应机理



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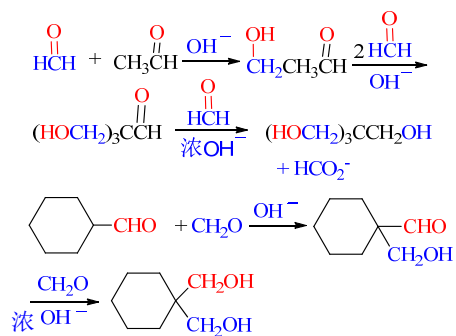
## 交叉的Cannizzarro 反应



OH<sup>-</sup>首先进攻正电性较强的羰基碳

48

## Example



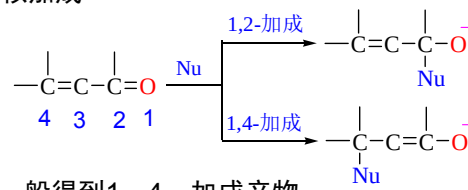
49

 $\alpha,\beta$ -不饱和醛酮

## 1. 亲电加成反应

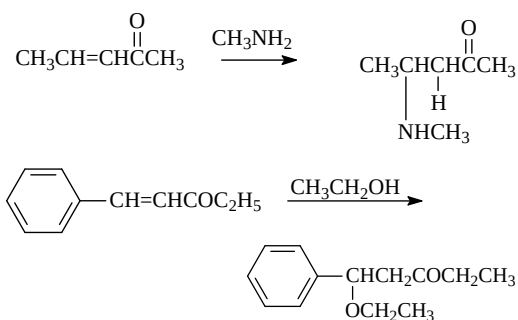


## 2. 亲核加成



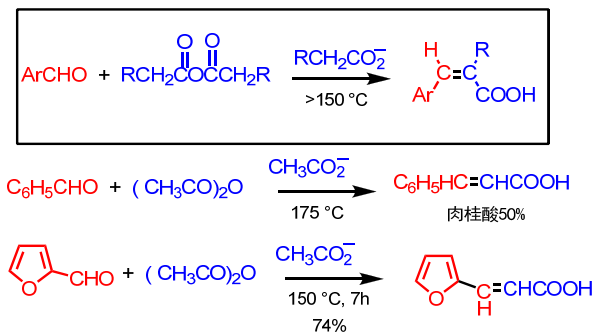
50

## Example



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## Perkin反应

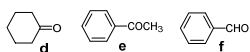
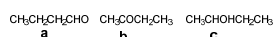


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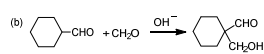
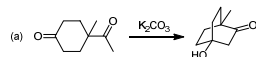


## 作业 (二)

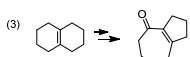
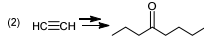
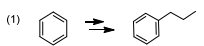
4. 用简单的化学方法鉴别下列6个化合物



5. 写出下列反应机理



6. 转化题



7. 以C2及以下有机物合成下列物质

